

UK Maths Trust

Junior Mathematical Olympiad

Tuesday 9 June 2026

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MARKETS

England & Wales: Year 8 or below | Scotland: S2 or below | Northern Ireland: Year 9 or below

Instructions

1. Do not open the paper until the invigilator tells you to do so.
2. Time allowed: **2 hours**.
3. The use of rulers, set squares and compasses is allowed, but **calculators and protractors are forbidden**. You are strongly encouraged to use geometrical instruments to construct large, accurate diagrams for geometry problems.
4. Start each question on an official answer sheet on which there is a **QR code**.
5. If you use additional sheets of (plain or lined) paper for a question, please write the following in the top left-hand corner of each sheet. (i) The question number. (ii) The page number for that question. (iii) The digits following the ‘:’ from the question’s answer sheet QR code.
Please do not write your name or initials on additional sheets.
6. **Write on one side of the paper only**. Make sure your writing and diagrams are clear and not too faint. (Your work will be scanned for marking.)
7. **Arrange your answer sheets in question order before they are collected**. If you are not submitting work for a particular problem, please remove the associated answer sheet.
8. **Your answers should be fully simplified, and exact**. They may contain symbols such as π , fractions, or square roots, if appropriate, but not decimal approximations.
9. For Question 1 only, some parts of the question are marked as answer-only. For these parts only, marks will be awarded for the numerical answer with no need for an explanation.
10. For all other questions, and other parts of Question 1, you should give full written solutions, including mathematical reasons as to why your method is correct and complete, including why no other solutions are possible. Just stating an answer, even a correct one, will earn you very few marks; also, incomplete or poorly presented solutions will not receive full marks.
11. **Scoring** Each question is worth up to 10 points. Answers can be awarded partial marks depending on the clarity of the participants mathematical presentation.
12. To accommodate candidates sitting in other time zones, please do not discuss the paper on the internet until **8am GMT on Thursday 11th June**, when the solutions video will be released at ukmt.org.uk/competitions-papers. Candidates in time zones more than 2 hours ahead of GMT must sit the paper on Wednesday 10th June (as defined locally).

Enquiries about the Junior Mathematical Olympiad should be sent to:

challenges@ukmt.org.uk www.ukmt.org.uk

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Concentrate firstly on one or two questions and then write out *full solutions* (not just brief ‘answers’), including mathematical reasons as to why your method is correct.

You will have done well if you hand in full solutions to two or more questions.

Do *not* hand in rough work.

1. In the following sum, the letters Y , U , M and S stand for single digits.

- (a) Find one possibility for the values of Y , U , M and S . Only the answer is required.

$$\begin{array}{r} Y \ U \ M \ S \\ + \ Y \ U \ M \\ \hline 2 \ 0 \ 2 \ 6 \end{array}$$

- (b) Prove that this is the only possible solution. A full explanation is required.

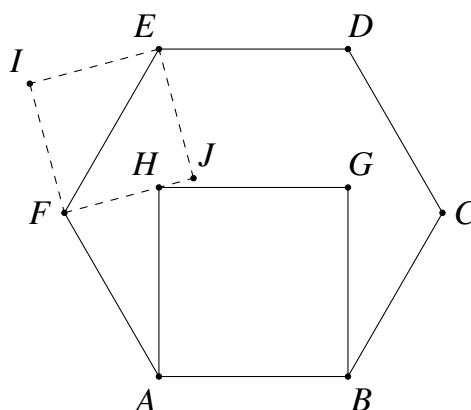
2. Prove that every number that can be written as the product of three consecutive positive integers can also be written as the sum of three consecutive positive integers.

3. Blackbeard and Brownbeard each have a pile of gold coins. One night, Blackbeard takes 3 gold coins from Brownbeard’s pile and adds them to his pile, and Brownbeard responds by taking half of the coins in Blackbeard’s pile and adding them to his pile. The next night, Brownbeard takes 3 coins from Blackbeard’s pile and adds them to his pile, and Blackbeard responds by taking half of the coins in Brownbeard’s pile and adding them to his pile. Both pirates now have the same number of gold coins as they had at the beginning.

Prove that the total number of gold coins is divisible by 3.

4. Suppose the vertices of a cube are labelled from 1 to 8. How many ways are there to choose three of these labels so that the corresponding three vertices are not all on the same face? *As this is an Olympiad problem, you must demonstrate that all possible choices are included in your count.*

5. $ABCDEF$ is a regular hexagon. The square $ABGH$ lies inside $ABCDEF$ and the vertex J of the square $EIFJ$ lies inside $ABCDEF$. Prove that H lies on the line FJ .



6. A stone is placed in the central square of a 3×3 grid. Each second, it moves one space vertically or horizontally to an adjacent square. After $2n$ seconds it is back to the centre. It is allowed to revisit cells that have been previously visited. Rotations and reflections of paths are considered different. In terms of n , how many possible paths could it have followed? For example, after 2 seconds there are 4 such paths.